Probabilistic Artificial Intelligence
Questions Pack

February 7, 2024

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. You must provide your answers to the questions on the accompanying **answer sheet** by blackening out the corresponding squares. Please carefully read the following instructions regarding the answer sheet:

- During the exam, you may use pencil and eraser to mark and edit your answers on the answer sheet. After the time is up, we will collect the questions pack. Make sure that you marked all your answers by then, at least with a pencil. After collecting the questions pack, we provide you with additional **10** minutes of time to blacken out the squares on the answer sheet with a **black pen**.
- As the answer sheet will be graded by a computer, please **make sure to blacken out the whole square. Do not use ticks or crosses**.
- Nothing written on the pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.
- Please make sure that your answer sheet is clean and do not write anything on the answer sheet except the squares you blackened out. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, questions award **1, 2, 3, or 4** points if answered correctly, depending on the difficulty of the question. You will encounter the following question types:

- **True or False.**
- **Multiple choice questions with a *single* correct answer.**

  These multiple choice questions have **exactly one** correct choice.
- **Multiple choice questions with *multiple* correct answers.**

  These multiple choice questions are marked with a "♣" and have **one or more** correct choices (at least one choice is correct). You will receive full points if you have chosen **all** the correct choices and **none** of the wrong choices. You will get zero points otherwise.

There are no negative points in this exam, so **0 points** are awarded if a question is answered wrongly or not attempted. The sections and questions are not ordered by difficulty. Thus, if you find a question too difficult, it may make sense to get back to it in the end.

Questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2.

Good luck!

1 Bayesian Linear Regression (13 points)

1.1 Hyperpriors (10 points)

We consider a dataset $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ of size n, where $\boldsymbol{x}_i \in \mathbb{R}^d$ denotes the feature vector and $y_i \in \mathbb{R}$ denotes the label of the i-th data point. Let ϵ_i be i.i.d. samples from the Gaussian distribution $\mathcal{N}(0, \lambda^{-1})$ for a given $\lambda > 0$. We collect the labels in a vector $\boldsymbol{y} \in \mathbb{R}^n$, the features in a matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d}$, and the noise in a vector $\boldsymbol{\epsilon} \in \mathbb{R}^n$. The labels are generated according to

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w} + \boldsymbol{\epsilon}.$$

To perform Bayesian Linear Regression, we consider the prior distribution over the parameter vector $\boldsymbol{w}$ to be $\mathcal{N}(\boldsymbol{\mu}, \lambda^{-1}\boldsymbol{I}_d)$, where $\boldsymbol{I}_d$ denotes the d-dimensional identity matrix and $\boldsymbol{\mu} \in \mathbb{R}^d$ is a hyperparameter.

Question 1 *(2 points)*
Given this Bayesian data model, what is the conditional covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \mathrm{Var}(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda)$?

A $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}\boldsymbol{I}_n$

■ $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}(\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n)$

C $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n$

D $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}(\frac{1}{n}\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n)$

Solution:
B. Let $\boldsymbol{f} = \boldsymbol{X}\boldsymbol{w}$. Since $\boldsymbol{w} \mid \boldsymbol{\mu}, \lambda \sim \mathcal{N}(\boldsymbol{\mu}, \lambda^{-1}\boldsymbol{I}_d)$, we see from lecture 1, slide 54 that

$$\boldsymbol{f} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\mu}, \lambda^{-1}\boldsymbol{X}\boldsymbol{X}^\top).$$

Thus,

$$\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\mu}, \lambda^{-1}\boldsymbol{X}\boldsymbol{X}^\top + \lambda^{-1}\boldsymbol{I}_n).$$

Question 2 *(3 points)*
Calculate the maximum likelihood estimate $\widehat{\boldsymbol{\mu}}_1$ of the hyperparameter $\boldsymbol{\mu}$.

A $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

B $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$

C $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

■ $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$

Solution:
D. We have

$$\widehat{\boldsymbol{\mu}}_1 = \arg\max_{\boldsymbol{\mu}} p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda) = \arg\max_{\boldsymbol{\mu}} \log \mathcal{N}(\boldsymbol{X}\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\boldsymbol{y}}).$$

We can simplify

$$\begin{aligned} -\log \mathcal{N}(\boldsymbol{X}\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\boldsymbol{y}}) &= \frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu})^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu}) + \text{const} \\ &= \frac{1}{2}\boldsymbol{\mu}^\top\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X}\boldsymbol{\mu} - \boldsymbol{\mu}^\top\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y} + \text{const}. \end{aligned}$$

Taking the gradient with respect to $\boldsymbol{\mu}$, we obtain

$$\nabla_{\boldsymbol{\mu}}\left(\frac{1}{2}\boldsymbol{\mu}^\top\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X}\boldsymbol{\mu} - \boldsymbol{y}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X}\boldsymbol{\mu} + \text{const}\right) = \boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X}\boldsymbol{\mu} - \boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}.$$

This is zero iff $\boldsymbol{\mu} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$.

Question 3 *(4 points)* Since we are unsure about the hyperparameter $\boldsymbol{\mu}$, we decide to model our uncertainty about $\boldsymbol{\mu}$ by placing the *hyperprior* $\boldsymbol{\mu} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$. Is the posterior distribution $p(\boldsymbol{\mu} \mid \boldsymbol{X}, \boldsymbol{y}, \lambda)$ a Gaussian distribution? If yes, what are its mean vector $\widehat{\boldsymbol{\mu}}_2$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\mu}}$?
Hint: For a Gaussian with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ it holds that

$$\log \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x} + \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \text{const.}$$

A It is not a Gaussian.
B $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$
■ $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$
D $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$
E $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$

Solution:
C. By Bayes' rule,

$$p(\boldsymbol{\mu} \mid \boldsymbol{X}, \boldsymbol{y}, \lambda) \propto p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda) \cdot p(\boldsymbol{\mu}).$$

Taking the negative logarithm,

$$-\log p(\boldsymbol{\mu} \mid \boldsymbol{X}, \boldsymbol{y}, \lambda) = -\log p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda) - \log p(\boldsymbol{\mu}) + \text{const.}$$

Analogously to the previous question, we can simplify

$$\begin{aligned} &= \frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu})^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu}) + \frac{1}{2}\boldsymbol{\mu}^\top\boldsymbol{\mu} + \text{const} \\ &= \frac{1}{2}\boldsymbol{\mu}^\top(\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)\boldsymbol{\mu} - \boldsymbol{\mu}^\top\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y} + \text{const.} \end{aligned}$$

This is a quadratic in $\boldsymbol{\mu}$, so the posterior distribution must be Gaussian. By matching the terms with the hint we obtain the covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$ and the mean vector $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$.

Question 4 *(1 point)*
We now want to make a prediction at a new input $\boldsymbol{x}^* \in \mathbb{R}^d$, for which the true label is $y^* = \boldsymbol{x}^{*\top}\boldsymbol{w} + \epsilon$. The posterior variance of y^* can be shown to be

$$\text{Var}\,(y^* \mid \boldsymbol{x}^*, \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) = \lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^* + \lambda^{-1}.$$

What are the aleatoric and epistemic components of the posterior variance?

A $\lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^*$ is the aleatoric and λ^{-1} the epistemic component
■ $\lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^*$ is the epistemic and λ^{-1} the aleatoric component

Solution:
B. The decomposition into aleatoric and epistemic uncertainty is discussed in lecture 2, slide 23.
In the following, we derive the posterior variance provided in the question. Analogously to the derivation of the posterior distribution of $\boldsymbol{\mu}$ (or remembering lecture 2, slide 18), we have

$$\text{Var}\,(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) = \lambda^{-1}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}.$$

Letting $f^* = \boldsymbol{x}^{*\top}\boldsymbol{w}$, we have $\text{Var}\,(f^* \mid \boldsymbol{x}^*, \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) = \lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^*$. Therefore,

$$\text{Var}\,(y^* \mid \boldsymbol{x}^*, \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) = \text{Var}\,(f^* \mid \boldsymbol{x}^*, \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) + \lambda^{-1}.$$

1.2 Short Questions (3 points)

Question 5 *(1 point)*
The MAP (Maximum A Posteriori) estimate of the weights in Bayesian linear regression with a Gaussian prior and Gaussian likelihood is the same as the mean of the posterior distribution over the weights.

■ True [B] False

Solution:
True. Under a Gaussian prior and likelihood, the posterior distribution is Gaussian (see lecture 2, slide 18). Since the mean of Gaussian is also its mode, the MAP estimate is the same as the mean of the posterior distribution.

Question 6 *(1 point)*
The Gaussian is the maximum entropy distribution among all distributions over $\mathbb{R}$ with a bounded mean.

[A] True ■ False

Solution:
False. The Gaussian is the maximum entropy distribution among all distributions over $\mathbb{R}$ with a given mean and variance (see homework 3, exercise 1). When no assumptions are made about the variance, the maximum entropy distribution is in general some Gibbs distribution.

Question 7 *(1 point)*
Bayesian linear regression can be implemented *online* where the data $(\boldsymbol{x} \in \mathbb{R}^d, y \in \mathbb{R})$ is coming one-by-one in an i.i.d. stream, and the posterior distribution is updated recursively.
Can the recursive update of a single round be performed in $\mathcal{O}(d)$ time?

[A] True ■ False

Solution:
False. In homework 1, exercise 5 you have seen that the recursive update of a single round can be performed in $\mathcal{O}(d^2)$ time. Storing the covariance matrix requires $\mathcal{O}(d^2)$ space.

2 Gaussian Processes (11 points)

For this section, we consider the data-generating process of $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$ given by

$$y_i = f(x_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where f is an unknown function and $\sigma > 0$ a known scalar. We model prior of the function f by a Gaussian process $f \sim \mathrm{GP}(0, k)$ with covariance function k.

2.1 Kernel Hyperparameters (4 points)

Suppose we observe the dataset $\mathcal{D} = \{(x_1, y_1), \ldots, (x_{10}, y_{10})\}$ and use the kernel

$$k(x, x') = \exp\left(-\frac{2\sin^2(\pi|x - x'|/p)}{l^2}\right) \exp\left(-\frac{(x - x')^2}{2l^2}\right),$$

where the length scale l and period length p are hyperparameters. Figure 1 shows the dataset $\mathcal{D}$ as well as the mean and 95% confidence region of the posterior $p(y|x, \mathcal{D}, l, p, \sigma^2)$ for four hyperparameter configurations:

- $l = 1, p = 0.4$
- $l = 1, p = 2$
- $l = 10, p = 0.4$
- $l = 10, p = 2$

Please match each of the plots in Figure 1 to the correct hyperparameter combination.

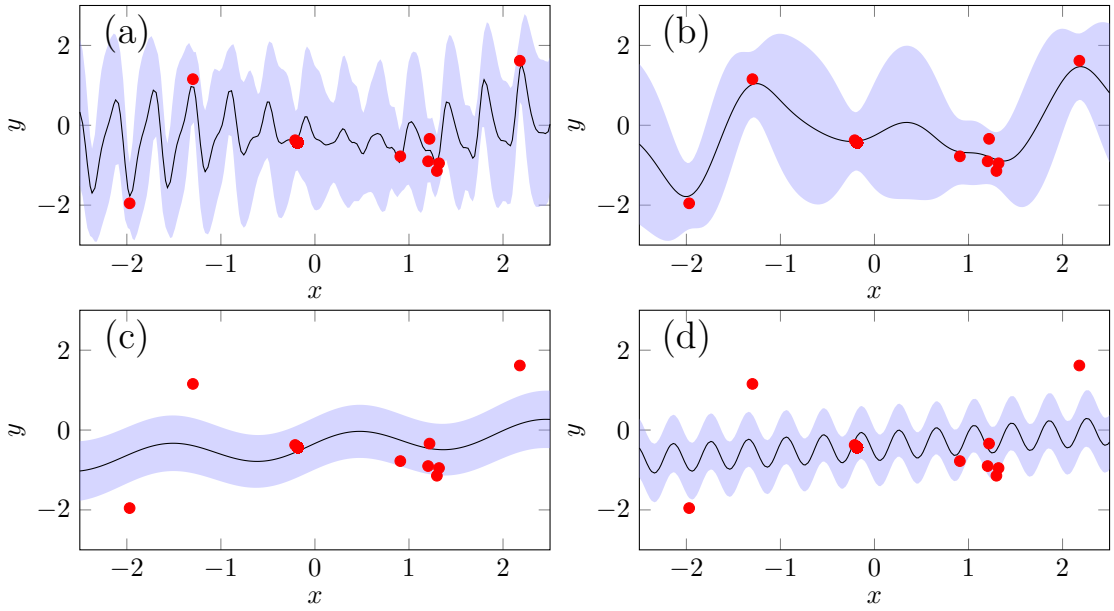


Figure 1

Question 8 *(1 points)* Which plot in Figure 1 corresponds to $l = 1$, $p = 0.4$?

■ (a) B (b) C (c) D (d)

Solution:
A. The solution of these questions is following from the Lecture 3 slides (pages 18 - 20), Homework 2 (exercise 1 c)), and the extension to the periodic kernel. In particular, p is comparatively small, such that the posterior mean will be exhibit a periodic behavior with high frequency. Since l is also comparatively small, this periodic behavior is only local, i.e., the amplitude of the periodic posterior mean will vary in the input domain. The only plot matching this description is (a).

Question 9 *(1 points)* Which plot in Figure 1 corresponds to $l = 1$, $p = 2$?

☒ (a) ☐ (b) ☐ (c) ☐ (d)

Solution:
B. Since p is relatively large and even larger than l, the periodic behavior will be almost invisible, such that the kernel behaves similar to a RBF kernel. Since (b) is the only plot without apparent periodic behavior, it fits this description best.

Question 10 *(1 points)* Which plot in Figure 1 corresponds to $l = 10$, $p = 0.4$?

☐ (a) ☐ (b) ☐ (c) ☒ (d)

Solution:
D. Since l is very large compared to the range of the input domain, the RBF component of kernel k will be almost constant. Therefore, the periodic behavior will be dominating in the mean function. As p is relatively small, the frequency of the periodic behavior will be high. This description is matched by plot (d).

Question 11 *(1 points)* Which plot in Figure 1 corresponds to $l = 10$, $p = 2$?

☐ (a) ☐ (b) ☒ (c) ☐ (d)

Solution:
C. Since l is very large compared to the range of the input domain, the RBF component of kernel k will be almost constant. Therefore, the periodic behavior will be dominating in the mean function. As p is comparatively large, the frequency of the periodic behavior will be low. This description is matched by plot (c).

2.2 Model Selection (4 points)

Question 12 *(4 points)*
Suppose we observe two data points $(x_1, y_1) = (0, 1)$ and $(x_2, y_2) = (1, 0)$ in the setting described at the top of this section. We now use the linear kernel

$$k(x, x') = xx'$$

to model the unknown function f. To tune the variance hyperparameter $\sigma^2 > 0$, we maximize the marginal likelihood $p(y_1, y_2|x_1, x_2, \sigma^2)$. Which of the following values for σ^2 maximizes $\log p(y_1, y_2|x_1, x_2, \sigma^2)$? You may assume that any $\sigma^2 > 0$ satisfying $\frac{d}{d\sigma^2} \log p(y_1, y_2|x_1, x_2, \sigma^2) = 0$ is a global maximum.

Hint: It holds that $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

A $\sigma^2 = \frac{1}{2}$ ■ $\sigma^2 = \frac{1}{\sqrt{2}}$ C $\sigma^2 = 1$ D $\sigma^2 = \sqrt{2}$ E $\sigma^2 = 2$

Solution:

B. We define the vectors $\boldsymbol{x} = (0, 1)^\top$ and $\boldsymbol{y} = (1, 0)^\top$ as well as kernel matrix $\boldsymbol{K}_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Following Lecture 4, slide 10, we have

$$\begin{aligned}
\log p(y_1, y_1 | x_1, x_2, \sigma^2) &= \log \mathcal{N}(\boldsymbol{y}; 0, \boldsymbol{K} + \sigma^2 \boldsymbol{I}) \\
&= \log \mathcal{N}(\boldsymbol{y}; 0, \boldsymbol{x}\boldsymbol{x}^\top + \sigma^2 \boldsymbol{I}) \\
&= -\frac{1}{2} \boldsymbol{y}^\top (\boldsymbol{x}\boldsymbol{x}^\top + \sigma^2 \boldsymbol{I})^{-1} \boldsymbol{y} - \frac{1}{2} \log \det(\boldsymbol{x}\boldsymbol{x}^\top + \sigma^2 \boldsymbol{I}) + \text{const.} \\
&= -\frac{1}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \sigma^2 & 0 \\ 0 & 1 + \sigma^2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{2} \log \det \begin{pmatrix} \sigma^2 & 0 \\ 0 & 1 + \sigma^2 \end{pmatrix} + \text{const.} \\
&= -\frac{1}{2\sigma^2} - \frac{1}{2} \log(\sigma^4 + \sigma^2) + \text{const.}
\end{aligned}$$

To find an extreme point for σ^2, we set the derivative with respect to σ^2 to zero. From the question description, this will be a global maximum. Let $\lambda = \sigma^2$. We then require

$$\begin{aligned}
\frac{d}{d\lambda} \log p(y_1, y_1 | x_1, x_2, \lambda) &= \frac{d}{d\lambda} \left(-\frac{1}{2\lambda} - \frac{1}{2} \log(\lambda^2 + \lambda) \right) \\
&= \frac{1}{2\lambda^2} - \frac{1}{2} \cdot \frac{2\lambda + 1}{\lambda^2 + \lambda} = 0
\end{aligned}$$

Thus,

$$\frac{1}{\lambda^2} = \frac{2\lambda + 1}{\lambda^2 + \lambda} \Leftrightarrow \lambda^2 + \lambda = 2\lambda^3 + \lambda^2 \Leftrightarrow \lambda = 2\lambda^3$$

Since $\lambda > 0$, we divide by λ and obtain $\lambda = \sigma^2 = 1/\sqrt{2}$.

2.3 Covariance Functions (3 points)

Question 13 *(1 point)*
The function $\exp(k(x, x'))$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function.

■ True B False

Solution:
True. The function $\exp(k(x, x'))$ is the exponential function with a kernel as its argument and therefore, a valid covariance function again (see Lecture 3 slides, page 22).

Question 14 *(1 point)*
The function $g(x, x') = k(ax, x')$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function and $a > 0$ is a constant.

A True ■ False

Solution:
False. The function $g(x, x') = k(ax, x')$ is not symmetric, so it is not a valid kernel (see Lecture 3 slides, page 22).

Question 15 *(1 point)*
The function $g(x, x') = k(x, x') - \frac{k(x,0)k(0,x')}{k(0,0)}$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function and $k(x, x) > 0$ for all $x \in \mathbb{R}$.

■ True B False

Solution:
True. The function $k(x, x') - \frac{k(x,0)k(0,x')}{k(0,0)}$ corresponds to the covariance function of a posterior Gaussian process with a single, noise-free data point (see, e.g., Lecture 4 slides, page 17). Thus, it is a valid covariance function.

3 Variational Inference and MCMC (17 Points)

3.1 Variational Inference (10 Points)

Question 16 *(1 point)*
Given a family of distributions $\mathcal{Q}$, the distribution $q^* \in \mathcal{Q}$ that minimizes $\mathrm{KL}(q \,\|\, p)$ for a distribution p is the same as the distribution that maximizes the ELBO for p.

■ True B False

Solution:
True. *(Markov Chain Monte Carlo, Slide 9).*

Question 17 *(2 points)*
We are given a dataset $\mathcal{D} = \{x_i\}_{i=1}^n$ sampled i.i.d. from a distribution $p(x \mid \theta)$ with unknown parameter θ. Assuming the prior $p(\theta)$, our goal is to approximate the posterior $p(\theta \mid \mathcal{D})$ of the unknown parameter with a distribution $q(\theta) \in \mathcal{Q}$, where $\mathcal{Q}$ is the set of all Gaussian distributions. Which of the following is necessarily true?

A If instead of maximizing the ELBO, we solve $\arg\max_{q \in \mathcal{Q}} \mathbb{E}_{\theta \sim q}[\log p(\mathcal{D} \mid \theta)]$, then the resulting distribution is the same as the distribution obtained via the Laplace approximation.

■ If instead of maximizing the ELBO, we solve $\arg\max_{q \in \mathcal{Q}} \mathbb{E}_{\theta \sim q}[\log p(\mathcal{D} \mid \theta)]$, then the resulting distribution is a Dirac delta at the maximum likelihood estimator for θ.

C The ELBO maximizer is the same as the Laplace approximation.

Solution:
B. *(Markov Chain Monte Carlo, Slide 9; Lecture Notes, Exercise 5.29)* The ELBO has two terms. What options A and B miss is the "closeness to the prior" term. This means that nothing stops the maximization from collapsing to a point mass. Indeed, one observes that the maximum is achieved by choosing q to be the Dirac delta located at $\arg\max_\theta \mathcal{D} \mid \theta)$. C is wrong, as Gaussian variational inference satisfies the conditions of the Laplace approximation *only on average* (see Eq. (5.71) in the Lecture Notes).

Question 18 *(3 points)*
Consider the family of distributions $p_\theta(x) = \mathcal{N}(x; \theta, \theta)$ on $\mathbb{R}$, that is, all Gaussian distributions with mean and variance equal to $\theta > 0$. We are given a differentiable function $f : \mathbb{R} \to \mathbb{R}$ and write f' for the derivative of f. Which of the following gives the correct expression for $\frac{d}{d\theta}\mathbb{E}_{x \sim p_\theta}[f(x)]$?

A $(\theta + \sqrt{\theta})\,\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(y)]$

B $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(\sqrt{\theta}\, y + \theta)]$

C $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(\theta y + \theta)]$

■ $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[(1 + \frac{y}{2\sqrt{\theta}}) f'(\sqrt{\theta}\, y + \theta)]$

E $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[(1 + y) f'(\theta y + \theta)]$

Solution:
D. *(Markov Chain Monte Carlo, Slide 12)* We shall use the reparametrization trick. Suppose $y \sim \mathcal{N}(0,1)$. Then $x = \sqrt{\theta}\, y + \theta$ would be distributed according to p_θ. Hence,

$$\begin{aligned}\frac{d}{d\theta}\mathbb{E}_{x\sim p_\theta}[f(x)] &= \frac{d}{d\theta}\mathbb{E}_{y\sim N(0,1)}[f(\sqrt{\theta}\, y+\theta)] \\ &= \mathbb{E}_{y\sim N(0,1)}\left[\frac{d}{d\theta}\{f(\sqrt{\theta}\, y+\theta)\}\right] \\ &= \mathbb{E}_{y\sim N(0,1)}\left[f'(\sqrt{\theta}\, y+\theta)\cdot\left(\frac{1}{2\sqrt{\theta}}y+1\right)\right].\end{aligned}$$

Question 19 *(4 points)*
We have a dataset $\{x_1, \ldots, x_n\}$ that contains i.i.d. samples from an unknown probability density p. Our goal is to train a neural network that approximates p. Let f_θ be the function defined by the neural network and consider the density $q_\theta(x) = \exp(-f_\theta(x))/Z(\theta)$, where $Z(\theta)$ is the normalization constant. Let $\{y_1, \ldots, y_m\}$ denote samples from q_θ. To approximate p, we use stochastic gradient descent to minimize the forward KL divergence $\mathrm{KL}(p \,\|\, q_\theta)$.
Which of the following is an unbiased estimator of $\nabla_\theta \mathrm{KL}(p \,\|\, q_\theta)$?

Hint: Write down the formula for the KL and compute the gradient analytically. Note that the normalization constant $Z(\theta)$ depends only on θ.

A $\frac{1}{n}\sum_{i=1}^n \exp(f_\theta(x_i))\nabla_\theta f_\theta(x_i)$

B $\frac{1}{n}\sum_{i=1}^n \exp(f_\theta(x_i))\nabla_\theta f_\theta(x_i) - \frac{1}{m}\sum_{j=1}^m \nabla_\theta f_\theta(y_j)$

C $\frac{1}{n}\sum_{i=1}^n \nabla_\theta f_\theta(x_i)$

■ $\frac{1}{n}\sum_{i=1}^n \nabla_\theta f_\theta(x_i) - \frac{1}{m}\sum_{j=1}^m \nabla_\theta f_\theta(y_j)$

E $\frac{1}{mn}\sum_{i=1}^n \sum_{j=1}^m \exp(f_\theta(y_j))\nabla_\theta f_\theta(x_i)$

Solution:
D. *(Markov Chain Monte Carlo, Slide 15)* We have by the definition of KL divergence

$$\mathrm{KL}(p \,\|\, q_\theta) = \int p(x) \log p(x)\, dx - \int p(x) \log q_\theta(x)\, dx$$

As the first term does not depend on θ, we have

$$\begin{aligned}
\nabla_\theta \mathrm{KL}(p \,\|\, q_\theta) &= -\nabla_\theta \int p(x) \log q_\theta(x)\, dx \\
&= -\nabla_\theta \int p(x)(-f_\theta(x) - \log Z(\theta))\, dx \\
&= \nabla_\theta \int p(x) f_\theta(x)\, dx + \nabla_\theta \log Z(\theta),
\end{aligned}$$

where in the last line we used the fact that p integrates to 1. The first term in the last equation is equal to

$$\int \nabla_\theta f_\theta(x)\, p(x)\, dx = \mathbb{E}_{x\sim p}[\nabla_\theta f_\theta(x)] \approx \frac{1}{n} \sum_{i=1}^{n} \nabla_\theta f_\theta(x_i).$$

For the second term, we have

$$\begin{aligned}
\nabla_\theta \log Z(\theta) &= \frac{\nabla_\theta Z(\theta)}{Z(\theta)} \\
&= \frac{1}{Z(\theta)} \nabla_\theta \int \exp(-f_\theta(x))\, dx \\
&= -\frac{1}{Z(\theta)} \int \exp(-f_\theta(x)) \nabla_\theta f_\theta(x)\, dx \\
&= -\int \frac{\exp(-f_\theta(x))}{Z(\theta)} \nabla_\theta f_\theta(x)\, dx \\
&= -\int q_\theta(x) \nabla_\theta f_\theta(x)\, dx \\
&= -\mathbb{E}_{x\sim q_\theta}[\nabla_\theta f_\theta(x)] \approx -\frac{1}{m} \sum_{j=1}^{m} \nabla_\theta f_\theta(y_j).
\end{aligned}$$

Putting the two together results in the correct answer.

3.2 MCMC and Sampling (7 Points)

Question 20 *(3 points)*
Consider a Markov chain $X_1, X_2, \ldots$ defined on a finite set $\mathcal{X}$ with transition probabilities $P(X_{k+1} = x' \mid X_k = x) = R(x' \mid x)$ for $k \geq 1$. Suppose that the distribution π is chosen in such a way that R and π satisfy the detailed balance property and assume that $X_1 \sim \pi$. Which of the following choices is correct?

A The chain is ergodic and therefore converges to the stationary distribution π.

B Regardless of the choice of the transition kernel R, the variables X_1 and X_2 are not independent.

■ $P(X_1 = x_1, \ldots, X_n = x_n) = P(X_1 = x_n, X_2 = x_{n-1}, \ldots, X_n = x_1)$ for any $x_1, \ldots, x_n \in \mathcal{X}$.

D For n large enough, we have $P(X_n = x) > 0$ for all $x \in \mathcal{X}$.

Solution:
C. *(Markov Chain Monte Carlo, Slide 32)* A is wrong: as ergodicity of the chain $X_1, X_2, \ldots$ is not implied by the detailed balance. Consider this counter example: let $\mathcal{X} = \{0, 1\}$, and set $R(1 \mid 1) = R(0 \mid 0) = 1$ and $\pi(0) = 1$. This chain satisfies the detailed balance property but is definitely not ergodic, as state 1 is not reachable from 0.
B is wrong: Take the following proposal: $R(x'|x) = \pi(x')$, that is, the transition probability does not depend on the state. This makes $X_1, X_2, \ldots$ independent copies of π.
C is correct: we show for $n = 3$ for simplicity:

$$\begin{aligned} P(X_1 = x_1, X_2 = x_2, X_3 = x_3) &= \pi(x_1)R(x_2 \mid x_1)R(x_3 \mid x_2) \\ &= R(x_1 \mid x_2)\pi(x_2)R(x_3 \mid x_2) \\ &= R(x_1 \mid x_2)R(x_2 \mid x_3)\pi(x_3) \\ &= P(X_1 = x_3, X_2 = x_2, X_3 = x_1). \end{aligned}$$

D is wrong: take the counter example of option A; $P(X_n = 1) = 0$ for all n.

Question 21 *(4 points)*
We want to sample from a truncated Gaussian distribution $\pi(x) \propto \exp(-x^2/2)\mathbb{I}(|x| < 1)$, where $\mathbb{I}(\cdot)$ is the indicator function. We use MCMC starting from the point 0. Which of the following algorithms has the correct stationary distribution π?

In each example, x denotes the current state of the chain, x' is the proposed state, and x^+ is the next state of the chain. All algorithms use the proposal distribution $R(x' \mid x) = \mathcal{N}(x'; (1-\gamma)x, 2\gamma)$, where $\gamma > 0$ is a step-size parameter.

A x^+ is x' if $x' \in (-1, 1)$ and otherwise is x.

B x^+ is x' if $x' \notin (-1, 1)$ and otherwise is x.

C x^+ is x' with probability $\min\left(1, \exp\left(\frac{-\gamma}{4}(x^2 - (x')^2)\right)\mathbb{I}(|x'| < 1)\right)$ and otherwise is x.

D x^+ is x' with probability $\min\left(1, \exp\left(\frac{\gamma}{4}(x^2 - (x')^2)\right)\right)$ and otherwise is x.

■ x^+ is x' with probability $\min\left(1, \exp\left(\frac{\gamma}{4}(x^2 - (x')^2)\right)\mathbb{I}(|x'| < 1)\right)$ and otherwise is x.

Solution:
E. *(Markov Chain Monte Carlo, Slides 45, 48)* First, observe that for the truncated Gaussian, we have $\frac{d}{dx}\log\pi(x) = -x$ for $x \in (-1,1)$. Therefore, a gradient step (as used in Langevin Monte Carlo or MALA) has the form $x - \gamma x = (1-\gamma)x$.
A is the usual Langevin Monte Carlo with a fixed step size and an accept/reject step that only concerns staying in the domain. The stationary distribution of Langevin with fixed step size is biased and is not π.
C is MALA for the target distribution $\mathcal{N}(0,1)$, and not the truncated Gaussian.
B and D only differ in the acceptance probability. We see below that the correct acceptance probability for MALA is D.
In a usual MALA implementation, the acceptance probability is $\min(1,\tilde{\alpha})$ where

$$\begin{aligned}
\tilde{\alpha} &= \frac{R(x \mid x')}{R(x' \mid x)} \cdot \frac{\pi(x')}{\pi(x)} \\
&= \exp\left(-\frac{(x-(1-\gamma)x')^2}{4\gamma} + \frac{(x'-(1-\gamma)x)^2}{4\gamma}\right) \cdot \exp\left(-\frac{(x')^2}{2} + \frac{x^2}{2}\right) \mathbb{I}(|x'|<1) \\
&= \exp\left(\frac{(1-(1-\gamma)^2)((x')^2-x^2)}{4\gamma}\right) \cdot \exp\left(-\frac{(x')^2}{2} + \frac{x^2}{2}\right) \mathbb{I}(|x'|<1) \\
&= \exp\left((\frac{1}{2}-\frac{\gamma}{4})((x')^2-x^2)\right) \cdot \exp\left(-\frac{(x')^2}{2} + \frac{x^2}{2}\right) \mathbb{I}(|x'|<1) \\
&= \exp\left(-\frac{\gamma}{4}((x')^2-x^2)\right) \mathbb{I}(|x'|<1)
\end{aligned}$$

4 Active Learning (12 points)

4.1 Mutual Information (4 points)

Let $p(x, y)$ be the joint probability density of the continuous random variables x and y defined on $\mathbb{R}$, respectively. Moreover, let

$$H(x|y) = -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log\left(\frac{p(x,y)}{p(y)}\right)dx\,dy$$
$$I(x;y) = H(x) - H(x|y)$$

be the conditional entropy and mutual information as defined in the lecture, respectively.

Question 22 *(1 point)*
If x and y are independent, then $I(x; y) = 0$.

■ True B False

Solution:
True. Was briefly mentioned in [Lecture 8, slide 8]. Otherwise, by definition, the mutual information $I(x; y)$ is

$$I(x;y) = H(x) - H(x|y)$$

It is clear that $I(x; y) = 0$ if and only if $H(x|y) = H(x)$. Expanding the definitions:

$$\begin{aligned} I(x;y) &= H(x) - H(x|y) \\ &= H(x) + H(y) - H(x,y) \\ &= \int_{\mathcal{X}}\int_{\mathcal{Y}} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}\,dx\,dy. \end{aligned}$$

Here we used $H(x|y) = H(x, y) - H(y)$ from [Lecture 8, slide 8]. If x and y are independent, then $p(x, y) = p(x)p(y)$ for all $x \in \mathcal{X}, y \in \mathcal{Y}$, and thus the logarithm term is zero, making $I(x; y) = 0$.

Question 23 ♣ *(3 points)* Which of the following statements are true? Mark all that apply.

A If $I(x; y) = 0$, then $H(y|x) = 0$.
B If $I(x; y) = 0$, then $H(x) = H(y)$.
C If $I(x; y) = 0$, then $H(y|x) = H(x)$.
■ If $I(x; y) = 0$, then $H(y|x) = H(y)$.

Solution:
D. As mentioned on [Lecture 8, slide 8], we have $I(x; y) = I(y; x) = H(y) - H(y|x)$. If $I(x; y) = 0$, then $H(y) = H(x|y)$. The other statements are false: take x to be close to a constant and y to be a distribution with high entropy. Concretely, take the marginal of x to be close to a Dirac delta centered at 0, and y to be $N(0, I)$, and let x and y be independent. Then $H(x) < 0$, $H(y) = \frac{d}{2}\log 2\pi e$, and $H(y|x) = \frac{d}{2}\log 2\pi e$.

4.2 Active Learning via Information Gain (8 points)

Our goal is to learn a linear function $f(\boldsymbol{x})$ defined on the unit circle $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_2 = 1\}$ from observations generated by

$$y_t = f(\boldsymbol{x}_t) + \epsilon_t,$$

with $\epsilon_t \sim \mathcal{N}(0,1)$. We perform Bayesian inference and place the prior $f \sim \mathrm{GP}(0,k)$ on the unknown function, where $k(\boldsymbol{x},\boldsymbol{x}') = \boldsymbol{x}^\top \boldsymbol{x}'$.

For a set of inputs $S = \{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_t\}$, we define the *information gain* of S as

$$F(S) := I(\boldsymbol{y}_{1:t}; f) = H(f) - H(f|\boldsymbol{y}_{1:t}),$$

where $H(f)$ is the entropy of f according to the GP prior and $H(f|\boldsymbol{y}_{1:t})$ is the conditional entropy after observing the outcomes $y_1,\ldots,y_t$. We define the *marginal information gain* $\Delta(\boldsymbol{x}|S)$ for a new point $\boldsymbol{x}$ as

$$\Delta(\boldsymbol{x}|S) = F(S \cup \{\boldsymbol{x}\}) - F(S).$$

Question 24 *(1 point)* Which of the following assertions is true?

■ $\Delta(\boldsymbol{x}|S) \geq 0$ for all S and $\boldsymbol{x} \in \mathcal{X}$.

B $\Delta(\boldsymbol{x}|S) \leq 0$ for all S and $\boldsymbol{x} \in \mathcal{X}$.

Solution:
A. Informally, information gain increases with more data. Formally, Lecture 8, slide 14 says

$$F(A \cup \{\boldsymbol{x}\}) - F(A) \geq F(B \cup \{\boldsymbol{x}\}) - F(B)$$

for all $A \subseteq B \subseteq D$ and $\boldsymbol{x} \in D$. Taking $A = S$ and $B = S \cup \{\boldsymbol{x}\}$ gives the desired result, since then

$$F(S \cup \{\boldsymbol{x}\}) - F(S) \geq F(B \cup \{\boldsymbol{x}\}) - F(B \cup \{\boldsymbol{x}\}) = 0.$$

Question 25 *(1 point)* Maximizing the differential information gain $\Delta(\boldsymbol{x}|S)$ is equivalent to picking the point $\boldsymbol{x} \in \mathcal{X}$ with maximum variance of the current posterior.

■ True B False

Solution:
True. See [Lecture 8, slide 11].

Question 26 *(2 point)* Which of the following is true *and* expresses that the function $S \mapsto F(S)$ is monotone submodular?

A For all S and S' such that $S \subseteq S'$, $F(S) \leq F(S')$.

B For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \leq \Delta(\boldsymbol{x}|S')$, for all $\boldsymbol{x} \in \mathcal{X}$.

■ For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \geq \Delta(\boldsymbol{x}|S')$, for all $\boldsymbol{x} \in \mathcal{X}$.

D For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \leq \Delta(\boldsymbol{x}'|S')$, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$.

E For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \geq \Delta(\boldsymbol{x}'|S')$, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$.

Solution:
C. Lecture 8, slide 13 defines the monotone submodularity of $F(S)$ as $F(A \cup \{\boldsymbol{x}\}) - F(A) \geq F(B \cup \{\boldsymbol{x}\}) - F(B)$ for all $A \subseteq B \subseteq D$ and $\boldsymbol{x} \in D$. This is equivalent to $\Delta(\boldsymbol{x}|A) \geq \Delta(\boldsymbol{x}|B)$ for all $A \subseteq B \subseteq D$ and $\boldsymbol{x} \in D$.

The other options are either not true or do not express that $F(S)$ is submodular. Option A implies that F is monotone, which is true, but it does not express that $F(S)$ is submodular. Option B is the monotone submodularity property in the reverse direction. To show that option D is false, let $S = S' = \{\mathbf{1}\}$ and choose two arbitrary vectors $\boldsymbol{x}, \boldsymbol{x'}$. Since D implies that both $F(\{\boldsymbol{x}, \mathbf{1}\}) \leq F(\{\boldsymbol{x}', \mathbf{1}\})$ and $F(\{\boldsymbol{x}', \mathbf{1}\}) \leq F(\{\boldsymbol{x}, \mathbf{1}\})$, we have $F(\{\boldsymbol{x}, \mathbf{1}\}) = F(\{\boldsymbol{x}', \mathbf{1}\})$, that is, the information gain is the same for all vectors $\boldsymbol{x}$. This is false and can be shown, for example, by computing the information gain for two choices of $\boldsymbol{x}, \boldsymbol{x'}$. In general, the information gain will not be the same, even for the linear kernel k. Option E allows the same argument as Option D.

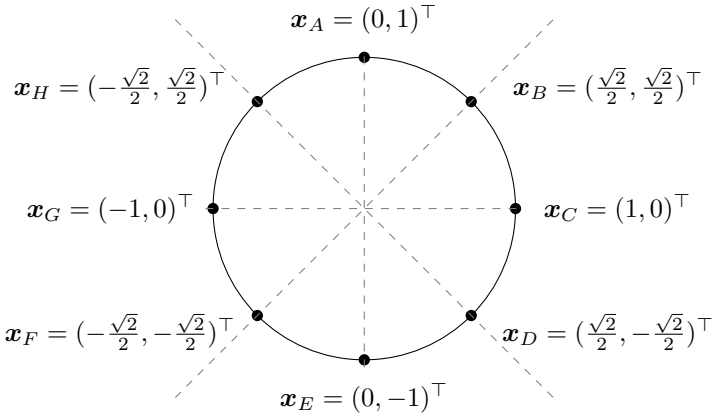


Figure 2: The unit ball in two dimensions. Points are symmetrically placed.

Question 27 ♣ *(4 points)*
Consider a greedy active learning algorithm that takes actions according to

$$\boldsymbol{x}_{t+1} \in \arg\max_{\boldsymbol{x} \in \mathcal{X}} \Delta(\boldsymbol{x}|S_t)$$

for all iterations $t \geq 1$ where $S_t = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t\}$ and recall that $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_2 = 1\}$. Suppose $\boldsymbol{x}_1 = \boldsymbol{x}_A = (0, 1)$ as shown in Figure 2. Which of the points shown in Figure 2 are valid as the next action $\boldsymbol{x}_2$ according to the criterion of the algorithm? Mark all that apply.

Hint: Note that $F(S) = \frac{1}{2} \log\det(\boldsymbol{I} + \boldsymbol{K}_S)$ where $\boldsymbol{I}$ is the identity matrix and $\boldsymbol{K}_S$ is the kernel matrix calculated for the points in S, that is, with entries $[\boldsymbol{K}_S]_{ij} := k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Moreover, it holds that

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

☐ A $\boldsymbol{x}_A$ ☐ B $\boldsymbol{x}_B$ ■ $\boldsymbol{x}_C$ ☐ D $\boldsymbol{x}_D$ ☐ E $\boldsymbol{x}_E$ ☐ F $\boldsymbol{x}_F$ ■ $\boldsymbol{x}_G$ ☐ H $\boldsymbol{x}_H$

Solution:
C, G. First, we note that

$$\arg\max_{\boldsymbol{x} \in \mathcal{X}} \Delta(\boldsymbol{x}|S_t) = \arg\max_{\boldsymbol{x} \in \mathcal{X}} F(S_t \cup \{\boldsymbol{x}\}) = \arg\max_{\boldsymbol{x}_2 \in \mathcal{X}} F(\{\boldsymbol{x}_1, \boldsymbol{x}_2\})$$

Then, by the hint,

$$F(\{\boldsymbol{x}_1, \boldsymbol{x}_2\}) = \frac{1}{2} \log\det \begin{pmatrix} 1 + \boldsymbol{x}_1^\top \boldsymbol{x}_1 & \boldsymbol{x}_1^\top \boldsymbol{x}_2 \\ \boldsymbol{x}_1^\top \boldsymbol{x}_2 & 1 + \boldsymbol{x}_2^\top \boldsymbol{x}_2 \end{pmatrix}$$

Set $\boldsymbol{x}_2 = (z_1, z_2)^\top$. Since $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$ and $\boldsymbol{x}_1 = (0, 1)^\top$ then,

$$F(\{\boldsymbol{x}_1, \boldsymbol{x}_2\}) = \frac{1}{2} \log\det \begin{pmatrix} 2 & z_2 \\ z_2 & 2 \end{pmatrix} = \frac{1}{2} \log(4 - z_2^2),$$

which is maximized if and only if $z_2 = 0$. Since $\boldsymbol{x}_2 \in \mathcal{X}$, then $z_1 = \pm 1$. Thus, the correct choices are $\boldsymbol{x}_C$ and $\boldsymbol{x}_G$ at $(\pm 1, 0)^\top$.

5 Reinforcement Learning (18 points)

5.1 Markov Decision Processes (9 points)

Your friend Ava is a professional runner, and you want to help her by designing a daily training plan. Your aim is to maximize the cumulative running form across T days. On a particular day, *running form* is the function $r(\alpha, \beta) = 1 + \beta - \alpha$ of the *fatigue* $\alpha \in \mathbb{R}_{\geq 0}$ and the *fitness* $\beta \in \mathbb{R}$. During each day, Ava can either *train* or *recover*. Your task is to design a policy that tells her what to do on each day t depending on her fatigue α_t and fitness β_t. Ava's initial fatigue is $\alpha_0 = 1$ and her initial fitness is $\beta_0 = 0$. We assume that fatigue and fitness follow the following deterministic dynamics:

$$\log \alpha_{t+1} = \begin{cases} \log \alpha_t + 1 & \text{when training on day } t \\ \log \alpha_t - 1 & \text{when recovering on day } t \end{cases}, \quad \beta_{t+1} = \begin{cases} \beta_t + (e-1) & \text{when training on day } t \\ \beta_t - (e-1) & \text{when recovering on day } t \end{cases}$$

Throughout this problem, you may assume for calculations that $e = 2.7$.

Question 28 *(1 point)*
Do the sequences of α_t and β_t (i.e., the dynamics of fatigue and fitness) satisfy the Markov assumption?

■ True B False

Solution:
True. See lecture 6, slide 27.

Question 29 *(2 points)*
Let $R_t = r(\alpha_t, \beta_t)$ be the running form on day t. A_t are the number of training days and B_t are the number of recovery days up to (not including) day t. The running form can be expressed as the following function of A_t and B_t only:

$$R_t = 1 + (e-1)(A_t - B_t) - e^{A_t - B_t}.$$

What is the maximum value for the running form on day t?

■ 0
B 5
C 4
D The running form is unbounded

Solution:
A. First, note that the running form on day t only depends on the difference $K_t := A_t - B_t$. The derivative of R_t with respect to K_t is $(e-1) - e^{K_t}$. Setting this to zero yields $K_t = \log(e-1) \approx 0.541$. Testing the two closest integers 0 and 1 shows that both are a maximum of the running form.

Question 30 *(3 points)*
The value of a policy π until day T is given by

$$J_T(\pi) = \mathbb{E}\left[\sum_{t=0}^{T-1} R_t\right].$$

Consider the policy π_1 according to which Ava trains each day. What is $J_3(\pi_1)$?

A $J_3(\pi_1) \approx -1$
■ $J_3(\pi_1) \approx -3$
B $J_3(\pi_1) \approx 0$
D $J_3(\pi_1) \approx -4$

Solution:
C. From the previous question and using $A_t - B_t = t$, as Ava trains each day, we can simplify

$$\begin{aligned} J_T(\pi_1) &= \sum_{t=0}^{T-1} 1 + (e-1)t - e^t \\ &= T + (e-1)\sum_{t=0}^{T-1} t - \sum_{t=0}^{T-1} e^t \\ &= T + (e-1)\frac{T(T-1)}{2} - \frac{1-e^T}{1-e} \\ &= T(1 + (e-1)(T-1)/2) - \frac{1-e^T}{1-e} \end{aligned}$$

where the third step follows from the formula for the sum of the first T integers and the formula for the sum of a geometric series. Plugging in $T = 3$ yields $J_3(\pi_1) \approx -2.952 \approx -3$.

Question 31 *(3 points)*
The class of all Markovian policies is $\Pi = \{\pi(\alpha_t, \beta_t) \in \{\text{train}, \text{recover}\}\}$ where $\pi(\alpha_t, \beta_t)$ determines whether to "train" or "recover" on day $t+1$. We say that a policy π is *optimal* if

$$\pi \in \arg\max_{\pi \in \Pi} J_3(\pi).$$

Consider the following family of policies $\widehat{\Pi} = \{\pi_p : p \in [0,1]\} \subseteq \Pi$ where π_p makes a decision independently of α_t, β_t:

$$\pi_p \equiv \begin{cases} \text{train} & \text{with probability } p \\ \text{recover} & \text{with probability } 1-p. \end{cases}$$

For example, according to the policy $\pi_{0.5}$, Ava would train or recover with probability $\frac{1}{2}$ each on every day. You now wonder which policy is optimal according to the definition of optimality given above. Which of the following statements is true?

A Policy π_1 is optimal.

B Policy $\pi_{0.5}$ is optimal.

C Policies $\pi_1, \pi_{0.5}$ are suboptimal, but there exists an optimal policy in $\widehat{\Pi}$.

■ $\widehat{\Pi}$ does not contain an optimal policy.

Solution:
D. Consider the policy

$$\pi^\star(K_t) = \begin{cases} \text{train} & \text{if } A_t - B_t = 0 \\ \text{recover} & \text{if } A_t - B_t = 1 \end{cases}$$

which is well-defined since under the deterministic dynamics and policy $\pi^\star$, $A_t - B_t \in \{0,1\}$, and $A_t - B_t$ can be computed from α_t and β_t (using the initial conditions). It is easy to see that $J_3(\pi^\star) = 0$ which is maximal since $R_t \leq 0$.
Thus, it remains to determine whether there exists some $\pi_p \in \widehat{\Pi}$ such that $J_3(\pi_p) = 0$. We make a case distinction: If $p < 1$, then with positive probability, the first action is to recover which implies $A_1 - B_1 = -1$ and $J_3(\pi_p) < 0$. Otherwise, $p = 1$, which is the policy to always train and also suboptimal as seen in the previous question. Hence, for all $\pi_p \in \widehat{\Pi}$, $J_3(\pi_p) < J_3(\pi^\star) = 0$.

5.2 Policy Gradient and Actor-Critic Methods (5 points)

Policy gradient methods learn a parameterized policy $\pi_{\boldsymbol{\theta}}$ by gradient ascent on the expected return

$$J(\boldsymbol{\theta}) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t\right].$$

The expectations in this subsection are always taken over trajectories realized under the policy π_θ. To perform gradient ascent, one needs to obtain an estimate of the gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$. In the lecture, you were introduced to the *policy gradient theorem*, which uses the *score function trick* to derive that

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}[Q(x,a) \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid x)]$$

where Q is the state-action value function of policy $\pi_{\boldsymbol{\theta}}$.

Question 32 *(1 point)*
For MDPs with a large state space and small action space, such as the MDP that models running form which was described in the previous section, policy gradient methods such as REINFORCE are computationally better suited than approximate Q-learning (such as DQN).

A True ■ False

Solution:
False. Approximate Q-learning methods such as DQN are suited precisely for the case where the state space is large while the action space is small. You were introduced to policy search methods in the context of scaling up to large (e.g., continuous) *action* spaces (see lecture 11, slide 27).

Question 33 *(2 points)*
Let $a \in \mathbb{R}$ and consider the stochastic policy

$$\pi_{\boldsymbol{\theta}}(a \mid x) = \mathcal{N}(a; f_{\boldsymbol{\theta}}(x), \sigma^2).$$

Starting from the policy gradient theorem, which expression does the policy gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ reduce to?

A $\mathbb{E}[Q(x,a)(f_{\boldsymbol{\theta}}(x) - a)\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)]$

B $\mathbb{E}[Q(x,a)(a - f_{\boldsymbol{\theta}}(x))\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)]$

C $\mathbb{E}\left[Q(x,a)\frac{(a - f_{\boldsymbol{\theta}}(x))}{2\sigma^2}\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)\right]$

■ $\mathbb{E}\left[Q(x,a)\frac{(a - f_{\boldsymbol{\theta}}(x))}{\sigma^2}\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)\right]$

Solution:
D. We need to compute $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid x)$. We have

$$\log \pi_{\boldsymbol{\theta}}(a \mid x) = \log \mathcal{N}(a; f_{\boldsymbol{\theta}}(x), \sigma^2) = -\frac{(f_{\boldsymbol{\theta}}(x) - a)^2}{2\sigma^2} + \text{const.}$$

By the chain rule,

$$\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid x) = \frac{(a - f_{\boldsymbol{\theta}}(x))}{\sigma^2}\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x).$$

Question 34 *(2 points)*
For the Gaussian policy from the previous question, we can use the reparameterization rather than the score function trick to obtain an unbiased estimate of the policy gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$.

A True ■ False

Solution:
False. No, we cannot use the reparameterization trick since we do not know how the states x depend on action a. These dependencies are determined by the unknown dynamics of the environment. The reparameterization trick would require reparameterization of this policy.

5.3 Short Questions (4 points)

Question 35 *(1 point)*
The multi-armed bandit problem is a special case of the reinforcement learning problem.

■ True B False

Solution:
True. The multi-armed bandit problem over a domain $\mathcal{X}$ coincides with reinforcement learning with a single fixed state and action set $\mathcal{X}$.

Question 36 *(1 point)*
An off-policy method can estimate the value function of a policy π by using data collected by following a different policy π'.

■ True B False

Solution:
True. This is the definition of an off-policy method, see lecture 10, slide 12.

Question 37 *(1 point)*
Recall that every MDP with finitely many states and actions possesses a deterministic optimal policy. Entropy regularization is used in reinforcement learning to penalize policies with high uncertainty, acting as a bias towards deterministic policies.

A True ■ False

Solution:
False. Entropy regularization is used to encourage the policy to exhibit uncertainty (i.e., have high entropy), see lecture 12, slide 39.

Question 38 *(1 point)*
Double DQN addresses the maximization bias of DQN (Deep Q-Network) by using in its loss $L(\boldsymbol{\theta})$ the target

$$r + \gamma Q(x', \arg\max_a Q(x', a; \boldsymbol{\theta}^{\text{old}}); \boldsymbol{\theta}^{\text{old}})$$

where the action a is selected with respect to the previous Q-network, rather than the target

$$r + \gamma Q(x', \arg\max_a Q(x', a; \boldsymbol{\theta}); \boldsymbol{\theta}^{\text{old}})$$

where a is selected with respect to the current Q-network.

A True ■ False

Solution:
False. It is the other way around. See lecture 11, slide 19.

CORRECTED

Answer Sheet of the Probabilistic Artificial Intelligence 2023/24 Exam

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

⟵ Please encode your student number on the left, and write your first and last names below.

Firstname and Lastname:

.....

Question 1: A ■ C D
Question 2: A B C ■
Question 3: A B ■ D E
Question 4: A ■
Question 5: ■ B
Question 6: A ■
Question 7: A ■
Question 8: ■ B C D
Question 9: A ■ C D
Question 10: A B C ■
Question 11: A B ■ D
Question 12: A ■ C D E
Question 13: ■ B
Question 14: A ■
Question 15: ■ B
Question 16: ■ B
Question 17: A ■ C
Question 18: A B C ■ E
Question 19: A B C ■ E
Question 20: A B ■ D
Question 21: A B C D ■
Question 22: ■ B
Question 23: A B C ■
Question 24: ■ B
Question 25: ■ B
Question 26: A B ■ D E
Question 27: A B ■ D E F ■ H
Question 28: ■ B
Question 29: ■ B C D
Question 30: A B ■ D
Question 31: A B C ■
Question 32: A ■
Question 33: A B C ■
Question 34: A ■
Question 35: ■ B
Question 36: ■ B
Question 37: A ■
Question 38: A ■